



Did you download a fake WhatsApp?

An official-looking version of WhatsApp appeared on the Google Play Store, and clocked more than a million downloads. <http://digit.in/FakeWhatsApp>



Sony's autonomous party bus

The SC-1 is outfitted with high-res image sensors, displays, AI, 5G data connectivity, alternative fuels and more onboard. <http://digit.in/SonyCar>

Let's face it

Facial recognition is being touted as the new password, but is the technology ready? The world of face recognition tech says yes.

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Looks matter. Now before we get into a debate of whether a person's looks play a crucial role in the way

they lead their life, from an aesthetic point of view, let us tell you that this one's about technology. It is no secret that the world of technology is headed towards a situation where the way you look is really important, all thanks to facial recognition and its growing ubiquity in everyday tech. While Apple's iPhone X is in no way the first application of facial recognition known to the tech world, it has, almost inevitably, thrown significant lime-light on the rising importance of facial recognition as the new password. But at this point, does the tech available truly satisfy the use cases we are implementing it for? Let's find out.

FACE DETECTION VS. FACE RECOGNITION

Before we explore what (and how) face recognition can do what it can do, it's important to understand the complexity of even the most basic face recognition systems. And the best way to do that is to compare it against a more rudimentary state of the technology – face detection. Detecting a face involves identifying the existence of certain parameters in an image – the eyes, a mouth, a nose etc. This identification involves looking for specific shapes that fit within a range of parameters. This technology has been around for quite a while – pick up any digital camera of the past decade and you'll probably see

face detection as a feature. In today's world, the ubiquitous face filters from Snapchat, Facebook, Instagram and more are examples of face detection in your everyday life.

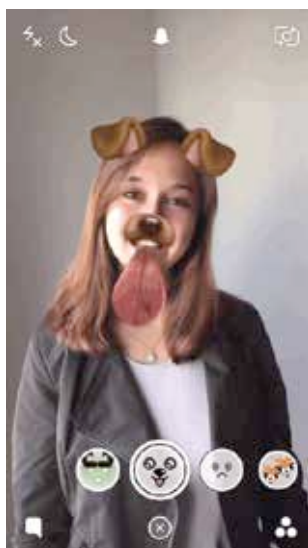
Recognition, on the other hand, casts a much smaller net. A face recognition system not only needs to detect a face, but also needs to know who that face belongs to. It could be matching the data against a database of employ-

ees, criminals, or even a single set that contains the parameters of the face of a phone owner. But in any of these cases, the median range of acceptance is a lot narrower. Which immediately establishes that face recognition cannot be performed by a digital system effectively unless images of a certain resolution are available. Thankfully, the tech in our everyday smartphones has reached a point where front cameras are powerful enough to provide such images. From this point forward it all depends upon the software.

ALGORITHMS

While there are a couple of different algorithms that are popularly used for face recognition, the overall process is somewhat similar across them and follows these steps, in order:

- Face detection: Done via methods like Haar-cascade classifier or History of Oriented Histograms.
- Face alignment: Using features like the eyes, the nose, etc.
- Appearance normalisation: Removing unnecessary details like colour, extra objects etc.
- Feature description: Parameters of the face are measured.



Snapchat brought face detection to the mainstream users